International Journal of Applied Earth Observation and Geoinformation 132 (2024) 103988

Contents lists available at ScienceDirect

International Journal of Applied Earth Observation and Geoinformation

journal homepage: www.elsevier.com/locate/jag

Enhancing Map Matching Accuracy Using Backtracking Technique

Hussein Alkhazraji [a,b,*], Triantafyllos Kanakis [a], Ali Al-Sherbaz [c], Wesam Sameer Bhaya [b], Michael Opoku Agyeman [a]

[a] Faculty of Arts, Science and Technology, University of Northampton, Waterside Campus, Northampton, NN1 5PH, United Kingdom
[b] College of Information Technology, University of Babylon, Babil, 51002, Iraq
[c] Institute of Continuing Education, University of Cambridge, Madingley, Cambridge, CB23 8AQ, United Kingdom

ARTICLE INFO

Dataset link: https://www.microsoft.com/en-us/research/publication/t-drive-trajectory-data-sample/

MSC:
0000
1111

Keywords:
GPS Trajectory
Map Matching
Backtracking Technique
T-drive Dataset

ABSTRACT

Map matching integrates positioning data with road network information to accurately identify a vehicle exact location on a specific road. Modern location-based services heavily rely on accurate map matching techniques to match observed trajectories with road networks. This paper presents a novel map matching approach that leverages topological information and integrates the backtracking technique to enhance the accuracy and robustness of map matching. By considering the logical relationships between road segments, our method achieves excellent results in scenarios with complex road geometries and challenging urban environments. We integrate the backtracking technique to address noise and ambiguity in trajectory data, resulting in improved precision and a reduced likelihood of false matches. Extensive experimentation using real-world trajectory datasets in Beijing validates the effectiveness of our approach, highlighting its superiority over traditional methods. The algorithm accuracy achieved as high as 99.92%. The results of the evaluation demonstrate that the proposed approach exhibits superior performance compared to earlier map-matching algorithms in both simple and complex road networks. Additionally, it can be effectively implemented on GPS datasets with different time intervals. In addition to the obvious map matching applications of our technology, it has the potential to be adapted across domains, with implications in transportation, urban planning, and location-based services. This paper not only presents a substantial advancement in map matching but also opens new avenues for location-based analyses enhancing and services across various domains.

1. Introduction

In contemporary times, taxi companies operating in urban areas have incorporated GPS devices into their vehicle fleets, enabling them to gain valuable insights (Satrinia and Saptawati, 2017). The data generated by these GPS devices consists of a series of sequential track points. When it comes to taxi trips, The collection of GPS data is a reasonably straightforward process that encompasses important details, including GPS Id, geographical information (longitude and latitude), and temporal information (timestamp). GPS data plays a crucial role in gathering traffic-related information. Compared to traditional sensors like loop detectors and camera monitoring systems, GPS data offers distinct advantages, including wider coverage and reduced costs associated with data collection and processing. The use of GPS data from vehicle trajectory data may serve several applications including land-use classification (Pan et al., 2012), measurement of travel time reliability (Wang et al., 2016), monitoring traffic congestion (Kong et al., 2015), detecting vehicle incidents (Dia and Thomas, 2011), and estimating traffic conditions (De Fabritiis et al., 2008; Sanaullah et al., 2016). Nevertheless, the presence of GPS errors resulting from device malfunctions, sensor inaccuracies, and environmental factors may introduce differences between the recorded GPS points and their true positions, thereby inducing spatial deviations. The map matching is the process of integrating positioning data with spatial road network data to accurately identify the specific road on which a vehicle is moving and detect the precise location of the vehicle along that road (Quddus et al., 2007). Consequently, the original data must undergo processing and map-matching procedures before being used on the road network.

Over the last three decades, map-matching algorithms have been taking into account, since the need for more GPS accuracy is exceed. algorithms evolved significantly, progressing from basic point-to-point matching techniques to sophisticated methodologies incorporating fuzzy logic theory. The fundamental and most straightforward strategy involves matching a given point to the closest road segment, commonly referred to as the "point-to-point" algorithm (Kong and Yang, 2019).

* Corresponding author at: Faculty of Arts, Science and Technology, University of Northampton, Waterside Campus, Northampton, NN1 5PH, United Kingdom.
E-mail address: hussein.alkhazraji@northampton.ac.uk (H. Alkhazraji).

https://doi.org/10.1016/j.jag.2024.103988
Received 23 December 2023; Received in revised form 29 May 2024; Accepted 16 June 2024
Available online 28 June 2024
1569-8432/© 2024 Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

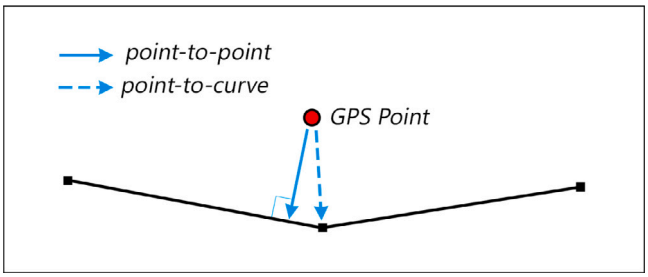

Fig. 1. Point-to-point and point-to-curve map-matching.

Another approach in Map Matching, known as the Point-to-Curve method, is often regarded as the most straightforward methodology. The process entails determining the anticipated location on the chosen road segment that has the smallest distance in relation to the original point. To ensure the selection of suitable road candidates, it is necessary for the distance between the original point and each road segment to adhere to a predetermined range. Although this strategy is simple, it has a limitation: it does not take into account the historical information of other points when estimating the projection point. Consequently, the selected projection point may not accurately reflect the intended location (Satrinia and Saptawati, 2017). The difference between Point-to-Point and Point-to-Curve methods described in Fig. 1.

The "Curve-to-Curve" method is an advancement built upon the Point-to-Curve method. It involves creating linear curves by connecting ordered points and then utilising the average distance between these curves to determine the corresponding road segment. The primary function of these algorithms is to match points by using geometric similarities. However, these approaches suffer from two main limitations. Firstly, as the number of points increases, the accuracy of the matching process declines rapidly. This is due to the possibility of the vehicle trajectory forming complex or intersecting polylines, making it challenging to accurately determine the road locations. Secondly, certain vehicle movements such as U-turns or crossing intersections can lead to erroneous conclusions during the map matching process (Bernstein et al., 1996).

Road connection is given key priority in topological techniques. The primary distinguishing feature of these methodologies is the ability to choose multiple pathways, situated either inside or crossing a designated buffer zone, prior to implementing the algorithm. Kong and Yang (2019). On the other hand, the shortest path method emphasises the route distance between two consecutive points as its main criterion. This method has demonstrated good performance in scenarios involving low-sampling-rate GPS data (Chen et al., 2014). However, it becomes inefficient when dealing with short intervals, as the computation of the shortest path requires a significant amount of time.

Weight-based methodologies use a scoring process that considers the weighted variables of distance, direction, and heading. These parameters are used to assess the probability of a match for any route within the collection of candidates. The road with the lowest score is subsequently identified as the correct choice (Velaga et al., 2009). Typically, the weighted parameters remain constant, which may not be suitable for all points. As a result, certain researchers opt for dynamic weighted parameters instead, allowing for more flexibility in the matching process (Hashemi and Karimi, 2016).

Advanced methodologies encompass a range of techniques such as the Kalman filter (Xu et al., 2010), fuzzy logic model (Yang et al., 2011), multiple hypothesis technique (Jabbour et al., 2008), hidden Markov model (Newson and Krumm, 2009), and other complex methodologies that use the principles of system dynamics theory or likelihood theory. These advanced methods typically achieve a high success rate in map matching, but they involve elaborate and computationally intensive calculations for all GPS points, even in cases where certain points lie within straightforward road segments and could be matched to the nearest road without extensive processing.

Typically, there exists a direct correlation between the accuracy of an algorithm and its associated operating costs. Presently, GPS solutions have achieved a horizontal accuracy of 3 m, with a 95% success rate (Washington, D.C., 2008). Meanwhile, it is commonly observed that the average distance between the two nearest roadways exceeds 20 m (Hoboken, 2006). Consequently, in the vast majority of cases, positioning points can be accurately matched to the nearest roads. It is only when the vehicle approaches distinctive traffic features such as intersections, parallel roads, or complex network that map matching becomes more challenging.

Several methods can be employed to enhance the accuracy of the map matching process, such as the utilisation of techniques like backtracking. The backtracking is a category of algorithms utilised to solve computational problems, specifically constraint satisfaction problems. These algorithms iteratively construct potential solutions and promptly discard a candidate if it is determined to be incapable of being completed as a valid solution. Gurari (1999). Thereby, this paper proposes a novel map matching algorithm based on topological information and backtracking technique, called Backtracking Technique Map Matching (BTMM). The key contributions of our research are as follows:

1. Developing a novel Backtracking Technique Map Matching (BTMM), by incorporating topological features of road networks, such as connectivity and directionality.
2. Integrating the backtracking technique into the map matching process. This enables the proposed method to handle ambiguous situations and correct earlier mapping decisions, yielding improved precision.
3. The algorithm outperforms different scenarios such as simple roads, parallel roads with opposite directions, complex networks, intersections, roundabout, U-turn.
4. Validating the effectiveness of our algorithm through extensive experimentation using real-world trajectory dataset.

The rest of the paper is organised as follows: The next Section 2 summarises the related work. Then, Section 3 will explain the framework of BTMM, such as concepts used in the proposed system. Section 3.2 describes the data preprocessing procedures used to prepare the data for the next Section 3.3 that will illustrate the BTMM algorithm. After that, Section 4 describes the experimental evaluation of map matching, The final Section 5 concludes the article.

2. Related work

In the field of map matching, researchers have developed many approaches to matching observed trajectories with digital road networks. This section presents an overview of key works within the field.

The research conducted by the work in Wei et al. (2013), provides insights into map-matching algorithms, classifying them into incremental max-weight, global max-weight, and global geometric methods. Max-weight techniques consider various parameters, including the alignment of recorded points and roads, distances between them, and the determination of the shortest route between recorded points, to identify the most accurate sequence. Specifically, incremental max-weight methods focus solely on previously recorded points for real-time usage, whereas global max-weight methods leverage the entire travel path for offline analysis, with the goal of improving mapping accuracy. While researchers in Kubicka et al. (2018) conduct a comprehensive categorisation of map-matching algorithms, recognising the nuanced challenges inherent in various application domains. Their approach aims to tailor solutions to the specific needs of different contexts, ranging from dynamic real-time navigation to more static offline surveying tasks. This classification framework emphasises the importance of adaptability in addressing the diverse requirements across these applications.

In a recent scholarly review, the work in Chao et al. (2020) critically examine existing classifications of map-matching algorithms, arguing that conventional frameworks no longer adequately capture the evolving scope of this field. In response, they propose a novel framework that not only considers the underlying algorithmic methodologies but also takes into account the specific application contexts in which these techniques are deployed. By combining insights from both algorithmic approaches and real-world application scenarios, their proposed framework seeks to provide a more holistic understanding of map-matching challenges and solutions. Moreover, their evaluation of various map-matching methods sheds light on the crucial role of sampling rates in algorithm performance. They highlight the delicate balance required to achieve optimal sampling frequencies, cautioning against both excessively high and low rates. Such insights underscore the importance of considering practical implementation factors alongside theoretical algorithmic considerations in the design and deployment of effective map-matching solutions.

In terms of developing incremental map matching algorithms, Zhu and Liu (2017) developed an incremental map matching technique that operates within the restrictions imposed by the road network layout. By thoroughly examining the geometric and topological limitations that impact map matching, The proposed technique demonstrates the capacity to mitigate the instability in matching caused by the low-frequency availability of GPS data and the computational complexities originating from extensive road networks inside urban areas. While Li et al. (2017) presented a road matching technique based on road tracking by using the road topology application. The algorithm adjusts its matching states and performs real-time matching adjustments based on variations in road topologies.

Also, Haoyan et al. (2023) introduced a map matching methodology called Path Increment Map Matching (PIMM). The authors of the study focused on the challenges associated with high-frequency trajectory data on complex road networks, including poor matching accuracy and sluggish matching speed. The suggested methodology consists of two distinct elements, namely integrated filtering and incremental matching. The road network is first optimised via the use of mixed filtering techniques. The subsequent procedure involves doing an incremental matching process by considering the paths as increments. The findings indicate that the proposed model achieved good matching results in terms of efficiency and projection accuracy in high-frequency data.

In recent times, there has been a surge of interest in various methodologies centred on dynamic weight. The algorithms in question extensively consider several forms of information to evaluate the likelihood of a match for each route within the candidate set (Zhu and Liu, 2017). In the research of Hashemi and Karimi (2016), a map matching approach proposed that incorporates the use of dynamic weight. The algorithm calculates the dynamic weight by taking into account many aspects, including the precision of the position, the velocity, and the distance travelled from the previous GPS locations. While in this study (Zhao et al., 2018) many factors were taken into account to assess the impact on GPS point matching. These factors included velocity, direction difference, vertical travel distance, and spatial relationship. The evaluation of the weight of each component was conducted using the Dempster–Shafer evidence theory. Nevertheless, in the context of the incremental matching algorithm, it is important to consider that incorrect matching results from prior points might potentially influence the matching process of subsequent points.

Depending on the "divide and conquer" idea, Liu et al. (2020) successfully differentiated the intersections from the road by using a method that included separating the track into intersecting and non-intersecting locations. This approach yielded excellent results in terms of achieving accurate matching findings. Also, Weber et al. (2010) proposed a method known as Selective Look-Ahead Map Matching (SLAMM) to tackle the issue of matching extensive and high-frequency data. The approach involves decomposing the trajectories into continuous segments, followed by doing map matching on a segment-by-segment basis. Yu et al. (2022) conducted significant enhancements to the performance and precision of the SLAMM algorithm via the optimisation of crucial sample selection and candidate arc selection. on the other hand, Kong and Yang (2019) introduced a scenario-based map-matching algorithm which offers distinct combinations of filters adapted for various operating locations. Filters used to check the GPS point location (intersection, parallel roads with same direction, parallel roads with opposite direction, ... etc.) and based on the determined filter the GPS point will be processed differently. The algorithm successfully produces great matching outcomes.

In addition, the study conducted by YIN et al. (2021) focused on addressing the issue of the topological matching method reliance on the initial matching position. To enhance the accuracy of determining the initial matching segment and initial location, Yin used a complete approach that included constraints related to speed, distance, and direction. The matching point is chosen based on the initial location, taking into account the constraints imposed by the speed and time data.

Table 1
Notations.

Symbol	Description
T_k	Trajectory of GPS records containing multiple P
P_i	Point of GPS, i is the point id
E_j	Edge or Road, j is the road id
$C(P_i)$	Candidate roads set for the Point P_i
C_j^i	Projection of the point P_i to the road E_j
$D(P_i, E_j)$	Distance between the point P_i and E_j
B	The circular buffer
B_{max}	The maximum buffer size allowed
B_{min}	The minimum buffer size allowed

3. BTMM framework

This section introduces an explanation regarding the proposed algorithm (BTMM). Fig. 2 illustrates the BTMM framework phases, including data preprocessing and map matching procedures.

3.1. Basic concepts

Definition 1. The dataset used consists of trajectories denoted by T_k where k is the trajectory id. Each trajectory contains multiple GPS points denoted by P_i, where i is the point id. Therefore, a trajectory T_k can be represented as $T_k = \{P_1, \dots, P_n\}$, where P_1 to P_n are the individual GPS points within the trajectory.

Definition 2. The roadmap network is denoted by a directed graph $G = (V, E)$, where G is the graph, V is a set of vertices, and E is a set of edges. Specifically, V is defined as $V = \{V_1, \dots, V_n\}$, and E defined as $E = \{E_1, \dots, E_m\}$, E_i where represents the directed edge from vertex V_i to vertex V_{i+1}.

Definition 3. The proposed system selects n roads around the GPS point P_i within a circular buffer B. These selected roads are denoted as the candidate roads set $C(P_i) = \{E_1, \dots, E_n\}$, where each E_i represents a candidate road within the buffer.

Definition 4. Each GPS point P_i will be matched to a road E_j, the projected point is denoted by C_j^i, Where C refer to the selected candidate road, i refer to the GPS point id and j refer to the road id.

Definition 5. The proposed system calculates the distance between the GPS point P_i and roads E_j, the distance is denoted by $D(P_i, E_j)$.

The symbols addressed in the definitions above, as well as the symbols to be used in this paper, are combined and presented in Table 1.

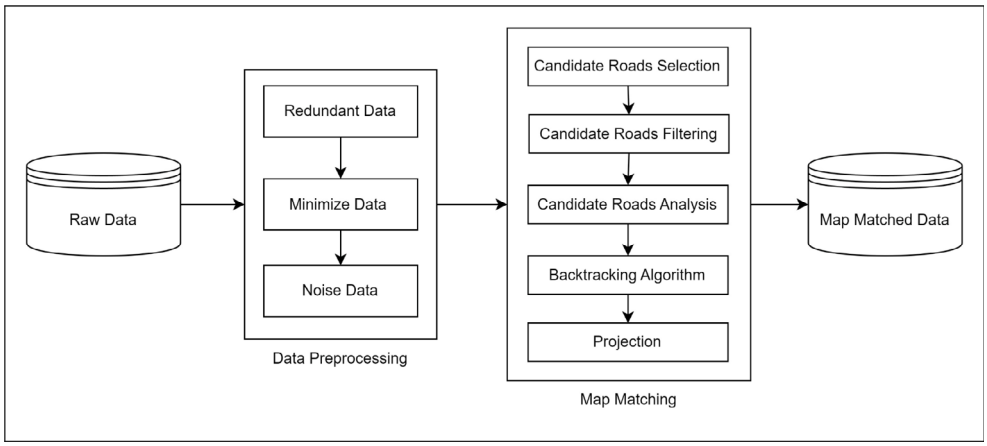

Fig. 2. The BTMM Framework.

3.2. Data preprocessing

The dataset mentioned in the data Section 4.1.1 requires a preprocessing procedure before to its application in the map-matching process. The data Preprocessing is an essential and sophisticated phase that plays a pivotal role in optimising datasets for subsequent tasks, such as map matching. The objective of data preprocessing is to meticulously eliminate irrelevant and erroneous data points, ensuring a refined and reliable dataset. Neglecting the data preprocessing can have detrimental effects on the outcome of subsequent analyses, primarily due to the presence of noisy data. This noise can severely hamper the efficiency of subsequent processes by consuming excessive computational resources and introducing errors into the system.

In the proposed system, A series of advanced data preprocessing steps are meticulously applied to guarantee the utmost data quality and integrity. These steps encompass a range of procedures as shown in Fig. 2, including the identification and removal of redundant or inconsistent data, and useless data. The dataset undergoes meticulous inspection to filter out any anomalies, outliers, or duplicate entries that could potentially skew the results of subsequent analyses.

3.2.1. Redundant data

In order to ensure data integrity and reliability, a first stage in the data preprocessing process entails removing redundant data that is included in the dataset. This step aims to identify and remove any duplicate or unnecessary information, ensuring that the dataset remains an organised structure and does not include any redundant entries.

This procedure involves iteratively examining the records of GPS points, comparing each current record to the subsequent record to determine their exact similarity in terms of taxi id, timestamp, latitude, and longitude. When two consecutive records exhibit identical attributes, the algorithm expertly eliminates one of the redundant records while retaining the other. As a consequence of this process, a significant total of 1,337,497 records have been successfully eliminated from the dataset, resulting in a refined and optimised data collection.

3.2.2. Minimise data

Another step in the preprocessing data is to remove repeated points that have exact location (latitude and longitude) and keep only the first and last point to know how long the vehicle stopped at that location. This method not only avoids loading a large amount of data into the memory, but also decreases the processing time thus increasing efficiency.

This process done by checking rows of GPS points, when the current row similar to the next row in terms of taxi id, latitude and longitude then the process keeps checking the subsequent records, if there are more than two rows identified with same location then all rows between the first and last rows will be removed. Fig. 3 shows a case for vehicle 950 stopped for a minute, if rows between the first row and last one are removed, it will still give the same situation for vehicle 950 which stopped for a minute. In result to this process 1,401,242 points have been removed from the dataset.

3.2.3. Noise data

Another crucial phase within the data preprocessing involves removing points that have unreasonable speed and consider them as "noise data". This kind of data has the most negative impact on the next processes. The speed can be determined by calculating travelled distance divided by travelled time.

To ensure that a point P_i is noise, the speed calculated with previous point P_{i-1} and next point P_{i+1}, if the two speeds higher than the max speed allowed in Beijing city (Gao et al., 2019) then the point (P_i) will be labelled as "noise data". For example, in Fig. 4, there is an instance of noisy data for a point where the distance between it and the previous point exceeds 9 kilometres, and the travel time between the two points is only 3 s. Such scenarios are highly improbable in real-world situations.

3.3. Map-matching algorithm

Map matching is an important process to correct GPS points that has inaccurate location and drift them to their actual location. The proposed approach is divided into five phases (as shown in Fig. 5): road selection for candidates, roads filtering, backtracking algorithm, analysis of candidate roads and projection. Each phase will be explained separately below.

3.3.1. Candidate roads selection

This is the first step applied to a GPS point P_i, it includes searching for roads locating within a circular buffer with the centre of point P_i, The success rate and computing time of map matching are influenced by the size of the buffer, which will be discussed in Section 4.1.2. Any road locating within or intersecting the buffer will be considered in the list of candidate roads $C(P_i)$ for the point P_i. In case the buffer does not include or intersect any roads, it will be expanded by a predetermined distance, referred to as the step distance, at every loop until the condition $C(P_i) \neq \emptyset$ is satisfied.

Taxi_id	Timestamp	Longitude	Latitude
950	02/02/2008 13:31:29	116.29459	39.88303
950	02/02/2008 13:31:34	116.29459	39.88303
950	02/02/2008 13:31:39	116.29459	39.88303
950	02/02/2008 13:31:44	116.29459	39.88303
950	02/02/2008 13:31:49	116.29459	39.88303
950	02/02/2008 13:31:59	116.29459	39.88303
950	02/02/2008 13:32:04	116.29459	39.88303
950	02/02/2008 13:32:09	116.29459	39.88303
950	02/02/2008 13:32:14	116.29459	39.88303
950	02/02/2008 13:32:19	116.29459	39.88303
950	02/02/2008 13:32:24	116.29459	39.88303
950	02/02/2008 13:32:29	116.29459	39.88303
950	02/02/2008 13:32:34	116.29459	39.88303
950	02/02/2008 13:32:39	116.29459	39.88303

Fig. 3. Sample for a vehicle stopped in the same location.

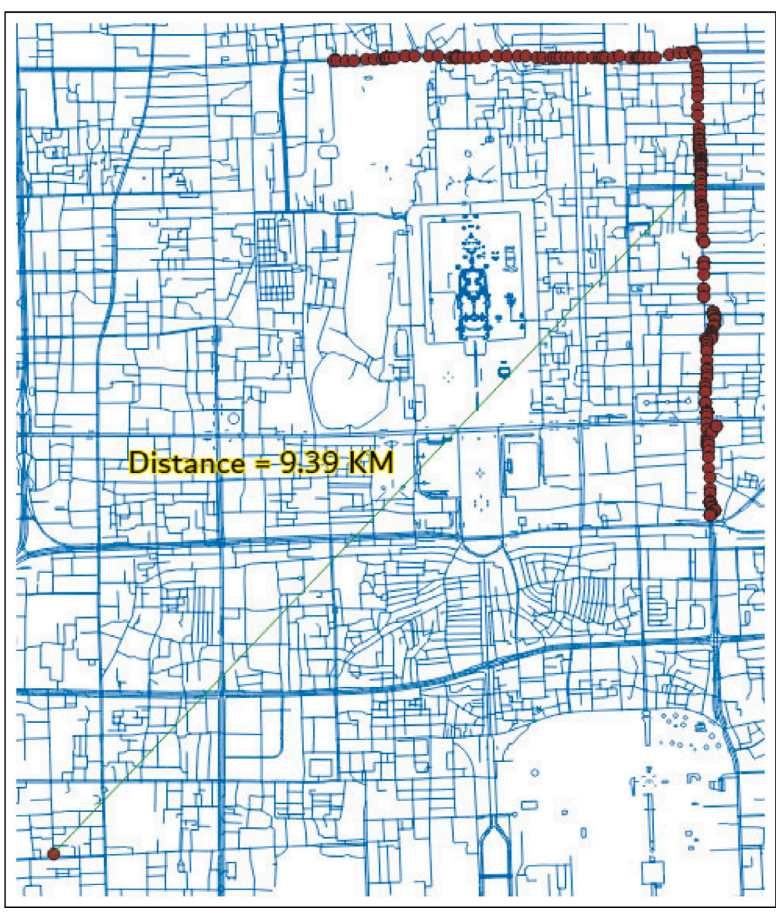

Fig. 4. Noise Data.

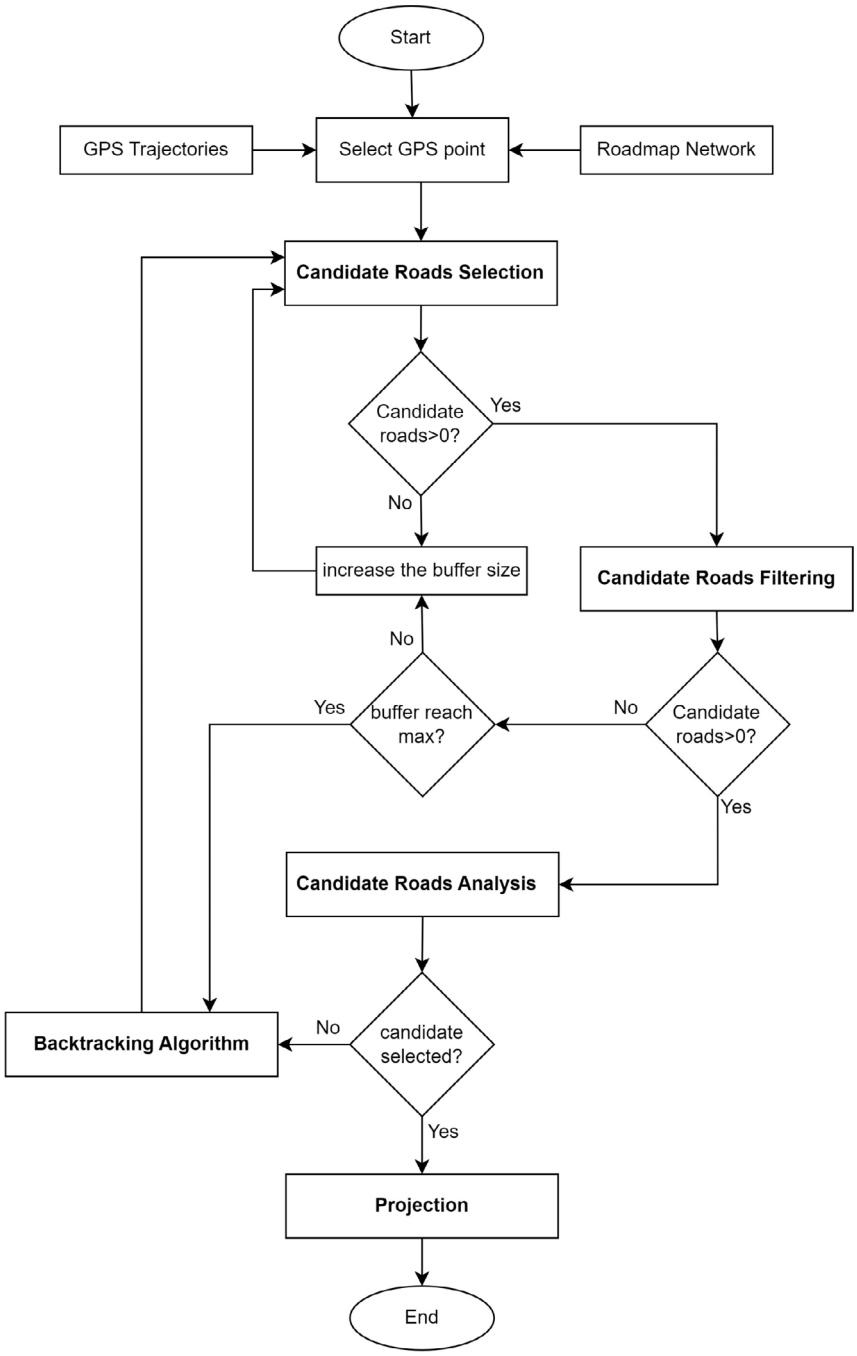

Fig. 5. The BTMM Flowchart.

3.3.2. *Candidate roads filtering*

In BTMM, the road to which P_i should be map-matched can be identified by the analysis of the topological connection between the previously matched road and the segments $C(P_i)$ and $C(P_{i+1})$. The computation of $C(P_i)$ and $C(P_{i+1})$ is performed for each P_i, while ensuring that the previous matching road is appropriately constructed.

It is important to mention that before the candidate roads filtering the proposed algorithm check if the nearest candidate road is the same as previous matched road, then P_i will be projected to that road. This is one of the steps that accelerate the map matching process.

In this procedure, the process will be divided into three steps. In the first step, the topological connectivity between $C(P_i)$ and previous matched road will be checked, any road not connected with previous matched road will be removed from $C(P_i)$. In the second step, the topological connectivity between $C(P_i)$ and $C(P_{i+1})$ will be checked by looping through each candidate road in the set $C(P_i)$, if the candidate is not connected to any candidate within the set $C(P_{i+1})$ then it will be removed from the $C(P_i)$. The last step, the candidate roads set $C(P_i)$ will be checked, if it contains a candidate road that excluded before by the backtracking algorithm, then it will be removed from the candidate roads set $C(P_i)$.

The proposed system will check only the new candidate roads after increasing the buffer size, and this is another step to save computational time and accelerate the map matching process. The process of increasing buffer size is limited until the buffer size reach the max size allowed by the proposed system which will be explained in Section 4.1.2.

The results of this procedure should be a filtered set of candidate roads $C(P_i)$. These candidate roads are either topologically connected

Table 2
Ordering Candidate Roads Set in Case A.

Candidate roads	Route	Distance to P_2
E_1	$E_1 - E_2 - E_3 - E_4$	3
E_2	$E_1 - E_2 - E_3 - E_4$	4
E_3	$E_1 - E_2 - E_3 - E_4$	7
E_4	$E_1 - E_2 - E_3 - E_4$	10
E_6	$E_1 - E_2 - E_3 - E_5 - E_6 - E_2 - E_3 - E_4$	1
E_5	$E_1 - E_2 - E_3 - E_5 - E_6 - E_2 - E_3 - E_4$	4

with previous matched roads and any candidate road of next point $C(P_{i+1})$, or not connected with previous matched roads but connected with any candidate road of subsequent point $C(P_{i+1})$. In the second case, the proposed system will remove any candidate road in a different direction. If the candidate roads set $C(P_i)$ still contains candidate roads then the nearest candidate road will be passed to the backtracking algorithm as a target road, otherwise the last matched road will be passed to the backtracking algorithm to be excluded.

3.3.3. Candidate roads analysis

In this important process, the candidate roads set $C(P_i)$ will be analysed to select the right road that P_i should be map-matched to. The analysis process will be as follows:

(1) If the candidate road set $C(P_i)$ contains one candidate road only, then it will be selected.

(2) If the candidate road set $C(P_i)$ contains more than one candidate road then these candidate roads will be arranged in ascending order based on the length of the route between the previous point P_{i-1} and the subsequent point P_{i+1}, passing through the candidate road. If more than one candidate road is located within the same route, then these candidate roads will be ordered ascendingly based on the distance between P_i and the candidate road, as shown in Table 2.

(3) The next step, the candidate road that is located within the shortest route and its distance to P_i is less than other candidates located within the shortest route will be selected. The selected candidate road must be the first element in the set $C(P_i)$ after the ordering process.

For example, Fig. 6 show three cases for P_2. the different between cases is that P_2 has different location in each case within an intersection. In all cases, the candidate roads set is $C(P_2) = \{E_1, E_2, E_3, E_4, E_5, E_6\}$, and the shortest route between the previous point P_1 and the next point P_3 is $E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_4$. The choice should be one of candidate roads $C(P_2)$ that located within the shortest route.

In case A, the candidate roads E_1, E_2, E_3, E_4 located within the shortest route, so the nearest one to P_2 between them is E_1. Although the candidate road $E6$ is the nearest candidate road to P_2, but it is not located within the shortest route.

In case B, the candidate road E_2 will be selected as its distance to P_2 is less than other candidates located within the shortest route. Also, the nearest candidate E_6 to P_2 will be ignored for the same reason in case A.

In case C, the nearest road is E_3 to P_2, and it locates within the shortest route. Then E_3 will be selected as the right candidate road to P_2.

(4) The final step in the analysis process is to confirm the choice from the previous step. this process will be done by checking if the selected road was not the nearest candidate to P_i, and the direction from P_{i-1} to P_i was same as the direction of the nearest candidate then the proposed system will pass the nearest candidate to the backtracking algorithm as a target road.

3.3.4. Projection

After the analysing process and selecting the candidate road that P_i should be map-matched to, the proposed system uses PostGIS Extension in the PostgreSQL database to call ST_ClosestPoint function in order to find the closest point located on the selected candidate road. In case the vertical projection of P_i falls on the road, it serves as the matched point (Fig. 7A). Otherwise, the matching point is determined as the endpoint of the segment (Fig. 7B).

3.3.5. Backtracking algorithm

The use of the backtracking approach serves to resolve incorrectly matched points and enhance the accuracy and effectiveness of BTMM. The backtracking technique will be used to either exclude a specific road from all previously associated points with that road, or correct a previous matched points based on a target road, as follows:

1. Excluding Roads: The backtracking algorithm excludes a specific road E_j from all previous points matched to that road, when the candidate roads set become empty $C(P_i) = \emptyset$. For instance, the buffer reaches max size and cannot find a candidate road to match P_i to. This case occurs in the candidate roads selection phase when the buffer size reaches max and cannot find a candidate road. Also, this case occurs in the candidate roads filtering phase when all candidates filtered out.

When this case occurs, the backtracking algorithm will check the selected road for previous point C_j^{i-1} and denoted that E_j must be excluded. The backtracking algorithm will loop through previous points and check if they match to the same road E_j, then E_j will be excluded from their candidate roads set until find a point matched to a different road.

The final step in this process, the backtracking algorithm will pass the last point Id to the map-matching algorithm to re-map matching them. for example, in Fig. 9 when the map matching continue filtering candidate road that are not connected with previous matched road E_2, and the $C(P_4) = \emptyset$, then, the backtracking will exclude E_2 from P_3 and P_2 then P_2 will pass to the map matching algorithm to re-map matching it. Hence, P_2 and P_3 will be matched to E_1 (on the end node). Consequently, P_4 will be matched to E_3.

2. Correct Matching Process: The second case, occurs in the candidate roads analysis phase when there is a candidate road nearer to P_i than the selected road to match P_i to, and its direction same as direction from P_{i-1} to P_i, then this candidate road will be the target road.

The backtracking algorithm will loop through previous points and correct their matching in case their matching road is not connected with the target road. For instance, in Fig. 8 the proposed system matched P_1 and P_2 to E_3 because it is the nearest. Hence, when the proposed system analyses the candidate roads $C(P_3)$ and it select E_4 to match P_3 to, but E_2 is nearer and its direction same as P_3 direction, then E_2 will be the target road. The backtracking algorithm starts looping through previous matched points, if P_2 is matched to E_3 which is not connected to the target road then P_2 will be matched to E_1 and E_1 will be the new target road. Also, the matching of P_1 will be corrected to E_1 and this correction process will continue as long as points matched to roads not connected to the target road, as shown in Fig. 11.

4. Experimental evaluation

This section present a case study of map matching in Beijing to validate and demonstrate the effectiveness of the proposed map matching method BTMM in matching observed trajectories with road networks.

4.1. Experimental setup

This section provides a details regarding dataset used in the conducted experiment. ALso, There are some parameters used in BTMM and have an impact on the success rate. These parameters are related to the buffer size and Directions Difference.

The evaluation takes place in a desktop PC on a 64-bit windows 10 operating system (12th Gen Intel(R) Core(TM) i7-12700F, 16 GB RAM). The algorithm is coded in MATLAB engine and PostgreSQL used to store data and to execute some queries through PostGIS extension which support some function such as calculating direction and distance. QGIS software used to display data (roads, GPS points, map matched points) from PostgreSQL database, all screenshots of the algorithm results were taken from the QGIS software.

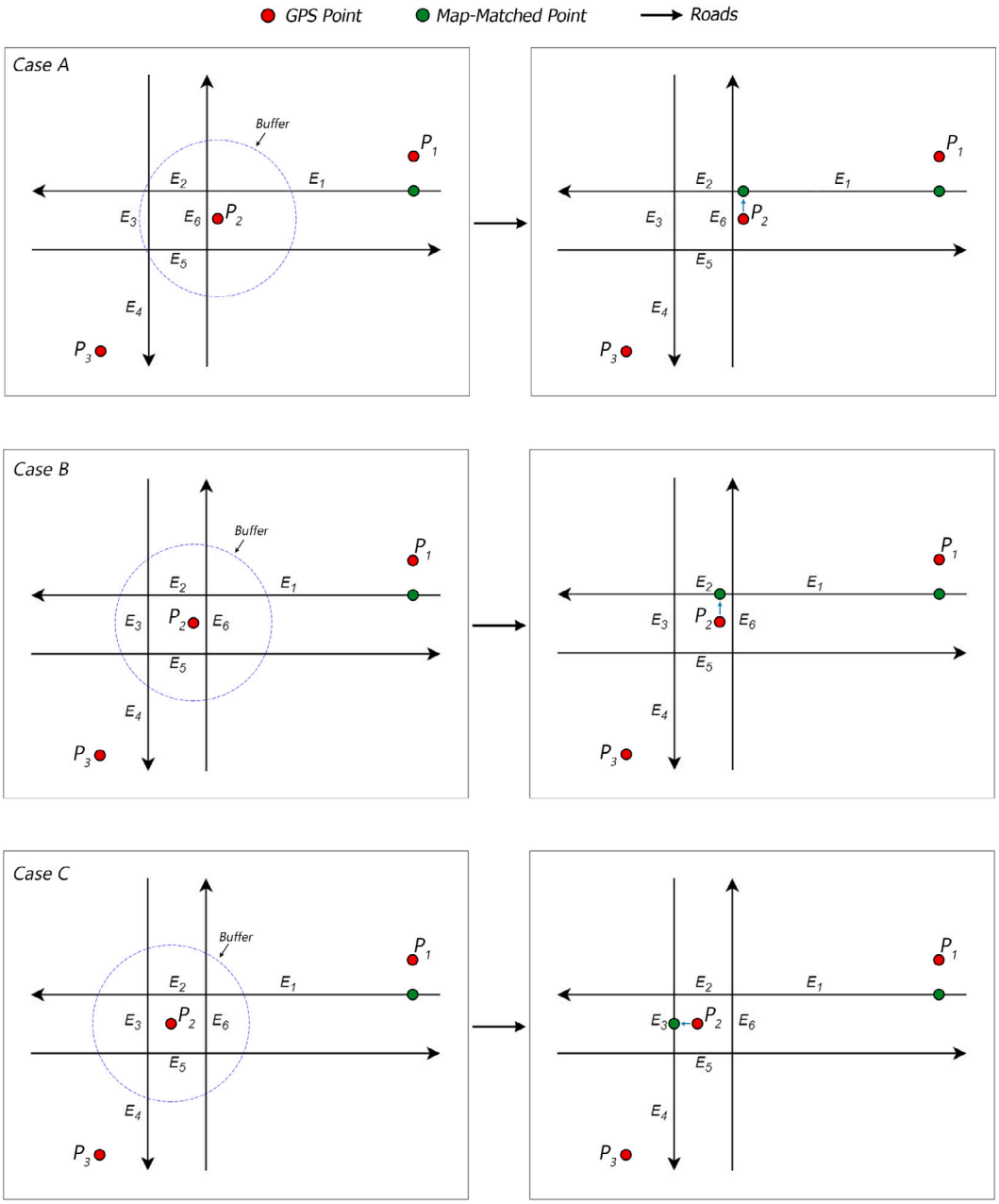

Fig. 6. Example of Matching GPS Point In Intersection.

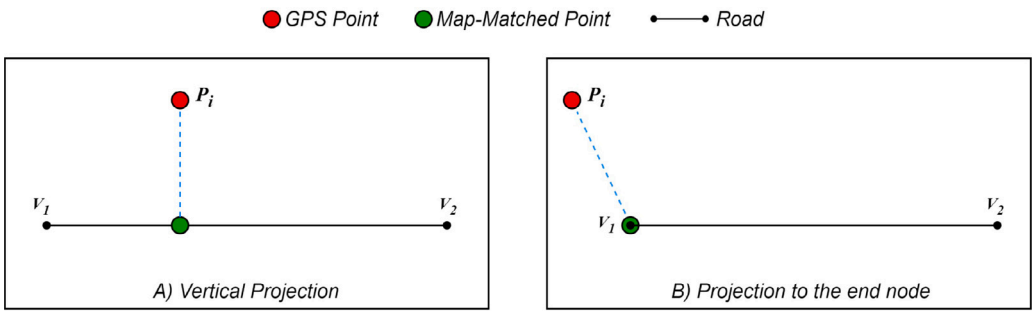

Fig. 7. Projection Process Cases.

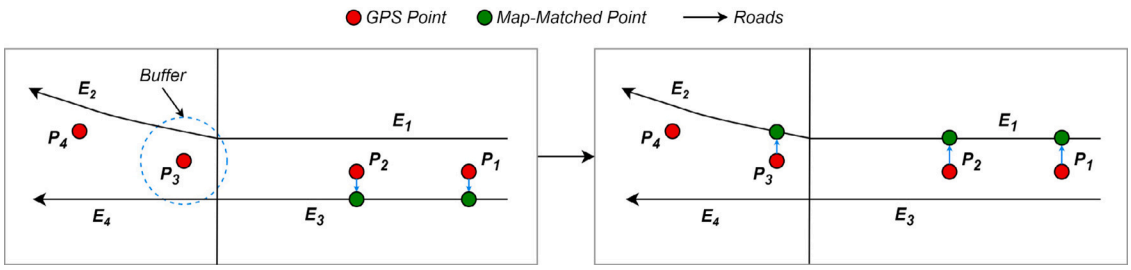

Fig. 8. Example of Correct Matching Process.

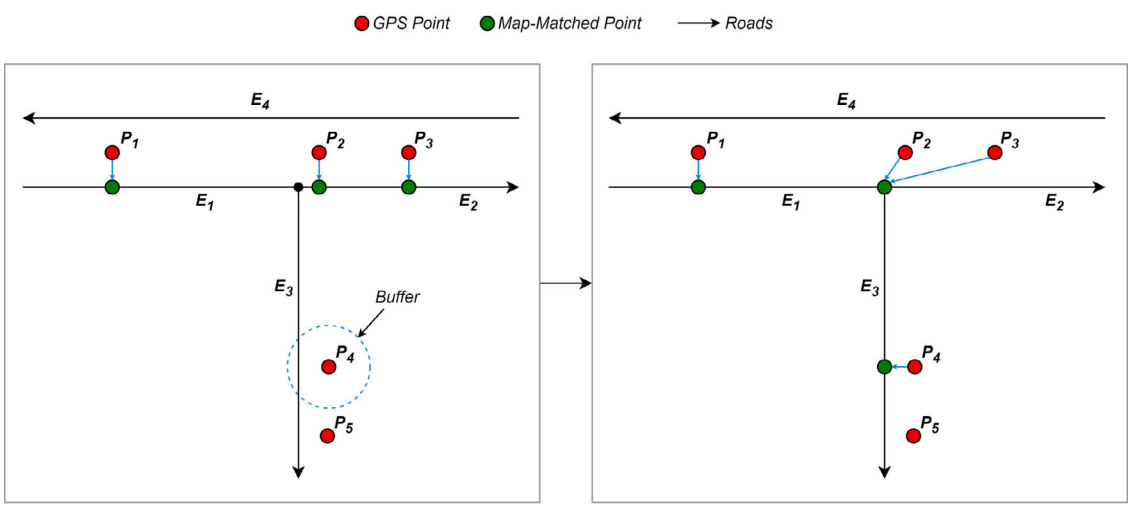

Fig. 9. Example of Excluding Roads.

4.1.1. Data

In the experiment, a two random trajectories (vehicles) were selected for one day period from the T-drive dataset (Zheng, 2011), which comprises a real-world GPS trajectories recorded over a span of one week from 10,357 taxis operating within the city of Beijing. The dataset contains an extensive collection of over 17 million points, with variations in the sampling rate ranging from 1 s to over 10 min per point.

The roadmap network for Beijing city was downloaded from OpenStreetMap website as XML file. It consists of 156,285 road segments. The XML file contains roads, buildings, traffic lights, ... etc. Only roads are exported and divided into segments. after that, road segments inserted into PostgreSQL database.

4.1.2. Buffer size

The selection of an appropriate buffer size is crucial in optimising the performance of BTMM. Increasing the buffer radius can result in longer running times due to the inclusion of unnecessary roads. Conversely, reducing the buffer radius might exclude accurate links or produce empty candidate road sets, necessitating an increase in buffer size and additional searches, which in turn consumes more time.

BTMM employs a variable buffer size instead of a static one, adapting to different scenarios. When a vehicle is moving within a specific area, the GPS points in this area generally have a consistent GPS signal strength. However, GPS signal strength might vary over time in the same area, such as when the vehicle is parked. For instance, if P_i is approximately 50 metres from candidate road E_j, then P_{i+1} is likely to be at a similar distance from E_j. Instead of starting the search with a static buffer size (e.g., 30 m) and potentially obtaining an empty candidate road set, the system dynamically adjusts the buffer size until the appropriate candidate road is found.

BTMM used a variable buffer size instead of static based on a scenario. The scenario considers when the vehicle is moving in a particular area, GPS points within this area have the same strength of GPS signal comparatively. GPS signal might change in the same area over time like when the vehicle parked. For instance, if P_i is near to a candidate road E_j by 50 metres then P_{i+1} will be slightly same distance from E_j, instead of start searching from static buffer size (e.g. 30 m) and getting empty candidate roads set and increasing the buffer size every time until find the right candidate road.

Consequently, the proposed system calculates the buffer size based on the original location of the last matched point P_{i-1} using Eq. (1). In cases where the calculated buffer B is less than 30 m, correct roads might not be selected. Therefore, the minimum buffer size is set to B_{min}, which is another parameter that contributes to the accuracy of the algorithm.

$$B = D(pi, C_j^i) - (D(pi, C_j^i) \mod 10) + 10 \tag{1}$$

The suggested scenario decrease the computational time, but it cannot guarantee all situations, like P_{i-1} located 25 metres from the selected candidate road E_j while P_i is located more than 50 metres from the nearest candidate road, in this case the proposed system will increase the buffer size until find a candidate road. The process of increasing buffer size by a distance called a step distance. In BTMM the step distance is set to 10 metres which improves the performance of the algorithm very well.

The process of increasing buffer size is limited until it reaches the max size, that means the buffer B should not exceed the max buffer size B_{max}. In BTMM B_{max} is set to 100 m. A relatively low percentage of GPS points were found to be located at that distance from the correct candidate road.

4.1.3. Directions difference

In the candidate road analysis, the proposed system calculates the directions of the candidate roads and compares them with the direction

of P_i. the proposed system will keep candidate that have the same direction of P_i. So, the difference in direction will effect on the success rate of the algorithm because low difference in direction will lead to remove correct candidate while high difference in direction will include unnecessary candidates. the proposed system performed very well and when it uses 45° as different degree in directions.

The difference in direction described in some literature (Kong and Yang, 2019) by using this equation $|\Theta_{road} - \Theta_{point}|$ which is not right. for example, if direction of $\Theta_{point} = 10°$, $\Theta_{road} = 350°$, then the difference will be 340°, while the actual difference is 20°. BTMM uses Eq. (2) to find the difference in direction.

$$\Theta_{difference} = 180 - |(|\Theta_{road} - \Theta_{point}|) - 180| \tag{2}$$

4.2. Evaluation metrics

In the assessment of our map matching approach, we rely on two primary evaluation metrics: accuracy and computational speed. These measurements offer a comprehensive understanding of our method performance and efficiency.

4.2.1. Accuracy

Accuracy serves as a fundamental indicator of the map matching quality achieved by BTMM, The measurement involves determining the ratio of accurately matched points to the overall number of points. The accuracy of map matching results is defined in Eq. (3).

$$\text{Accuracy} = \frac{N_{Correct}}{N_{TotalPoints}} \times 100 \tag{3}$$

Where $N_{Correct}$ represents the number of GPS points that have been correctly matched to their corresponding road segments. $N_{TotalPoints}$ is the total number of GPS points in the evaluation dataset.

The use of accuracy providing insights into the overall performance of BTMM. A higher accuracy score indicates a more effective matching process, while a lower accuracy score suggests potential challenges in the map matching process.

4.2.2. Computational speed

Computational speed is another metric to assess the efficiency of our map matching algorithm. It calculates the amount of time needed to process the trajectory data and finish the map matching assignment.

The computational speed of map matching results is defined as the number of points processed per second (PPS), as shown in Eq. (4).

$$\text{Computational Speed (PPS)} = \frac{N_{MatchedPoints}}{Time(s)} \tag{4}$$

Where $N_{MatchedPoints}$ represents the number of matched points to their corresponding road segments. $Time(s)$ is the total time taken to process the points. A higher PPS value indicates a more efficient algorithm capable of processing large datasets quickly.

4.3. Baseline methods

In this research, we conducted comparative evaluations of our proposed Backtracking-based Map Matching (BTMM) algorithm with three existing map matching methods, namely, the scenario-based map-matching algorithm (Kong and Yang, 2019), the MDP-MM algorithm (Chen et al., 2014), and the dynamic weight-based algorithm (Hashemi and Karimi, 2016). These selections were made based on their relevance to our research goals and their distinctive characteristics.

1. **Scenario-Based Algorithm:** The scenario-based algorithm is chosen due to its adaptability to different scenarios within road networks. It employs a combination of five filters to handle various situations effectively. We include this algorithm in our comparative evaluation to understand how our method performs in scenarios that demand adaptability and robustness, as our approach leverages topological insights to address similar challenges.
2. **MDP-MM Algorithm:** The MDP-MM algorithm is chosen for its unique approach to map matching. It employs a weighted combination of travel distance and perpendicular distance as indications in order determine the right candidate. This method offers a different perspective from traditional geometric-based map matching and serves as an interesting point of comparison with our topological and backtracking approach.
3. **Dynamic Weight-Based Algorithm:** The dynamic weight-based algorithm has garnered significant attention in recent research. The proposed approach demonstrates innovation in using dynamic weights to detect junctions and choose the optimal road based on factors such as proximity, direction difference, and heading difference. Given its relevance to modern map matching challenges, we include it as a baseline to assess how our topological information and backtracking strategy complement or surpass its capabilities.

Each of these baseline methods represents a distinct aspect of map matching, and their inclusion allows us to comprehensively assess the strengths and weaknesses of our BTMM algorithm in various contexts. The comparative evaluation results, presented in the following section, provide insights into the efficacy of our novel approach in enhancing map matching accuracy and robustness.

4.4. Experimental results

This part comprehensively examines accuracy scores and analysis for each experimental scenario, providing insights into the efficacy of our strategy under different conditions. The selected trajectories points were labelled with the corresponding road, hence facilitating fast comparison and calculation of the outcome. The success rate of our proposed algorithm BTMM under different scenarios (Simple roads, Parallel roads, Intersections, U-turn, roundabout, anomaly cases) is as high as 99.92%.

The BTMM algorithm evaluated on different environments such as complex networks, Parallel roads with opposite directions, U-turn Roads, Intersections, and roundabouts as shown in Fig. 10.

The task of U-turn map-matching is a significant challenge for the majority of algorithms. As an example, U-turns constitute around 10% of the instances when the Multiple Hypothesis Technique (MHT) algorithm fails to provide a successful match (Schuessler and Axhausen, 2009). Fig. 10C shows the result of evaluating BTMM on U-turns. The process of map matching in the area of intersections has posed challenges for several methods (Velaga et al., 2012). BTMM can overcome different kinds of intersections by using a combination of topological features and backtracking techniques.

Most algorithms take decision for P_i to be map matched to a candidate road E_j once, while BTMM algorithm may change the decision in case it identifies the selected candidate road is not correct. Fig. 11 shows the significant role of the backtracking technique in correcting errors in matching points.

Additionally, we compare the performance of our proposed Backtracking-based Map Matching (BTMM) algorithm against three selected baseline methods, namely, the MDP-MM algorithm, the dynamic weight-based algorithm, and the scenario-based map-matching algorithm, under various sampling intervals.

Fig. 12 illustrates The evaluation of these algorithms at various sample intervals. Notably, BTMM demonstrates excellent map matching accuracy, consistently outperforming the baseline methods and achieving the highest success rate.

BTMM has outperform other algorithms and achieve the best success rate. BTMM accuracy was slightly higher than the scenario-based algorithm with 2.00%. Also, BTMM success rate was 4.00% and 6.00% higher than dynamic weight-based and MDP-MM respectively. These results underscore the effectiveness of BTMM in correctly matching trajectories to road segments.

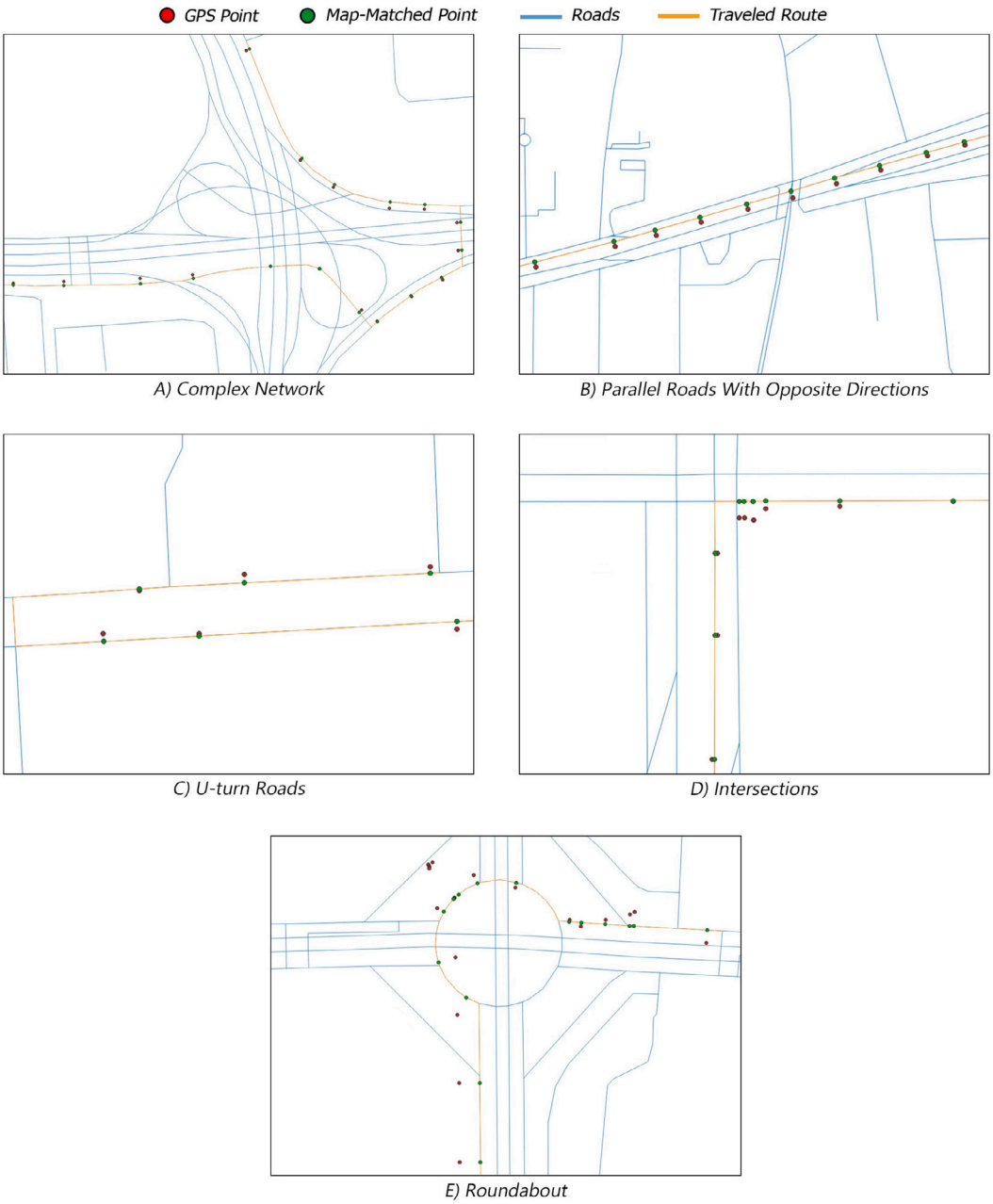

Fig. 10. The BTMM Results.

We also compared the performance of BTMM in terms of computational speed. The BTMM algorithm processes data 1.6 times faster than the MDP-MM algorithm, 1.4 times faster than the dynamic weight-based algorithm, and 1.2 times faster than the scenario-based map-matching algorithm, as shown in Fig. 13.

4.5. Discussion

This section provides an analysis of the experimental evaluation, focusing on the strengths of our BTMM algorithm compared to the baseline methods. These strengths are attributed to specific enhancements incorporated into BTMM, hence setting it apart from its competitors.

One of the primary strengths of BTMM algorithm lies in its use of the backtracking technique as explained in the results section. Additionally, employ dynamic buffer size unlike traditional methods with constant buffer sizes (as illustrated in Buffer Size section). This dynamic approach significantly contributes to the higher success rate achieved, as it allows for more flexible and context-aware in map matching process.

To further elaborate on the impact of buffer size on the system's performance, we conducted several experiments to analyse different buffer sizes under various conditions. The buffer sizes were categorised as follows: small (20-40 m), medium (40-60 m), and large (greater than 60 m). The results demonstrate that larger buffer sizes tend to increase computation time due to the inclusion of more candidate roads, while smaller buffer sizes may lead to higher rates of exclusion errors, necessitating additional searches and adjustments. Importantly, the accuracy of the system remains consistent across different buffer sizes because the backtracking technique corrects any initial errors in road selection. By employing a variable (dynamic) buffer size strategy, the system adapts to changes in the environment and maintains a balance between computational efficiency and accuracy. For instance, in areas with dense road networks, a larger buffer size ensures that all potential candidate roads are considered, improving the likelihood of correct matches on the first attempt. Conversely, in sparse road networks, a smaller buffer size reduces unnecessary computations.

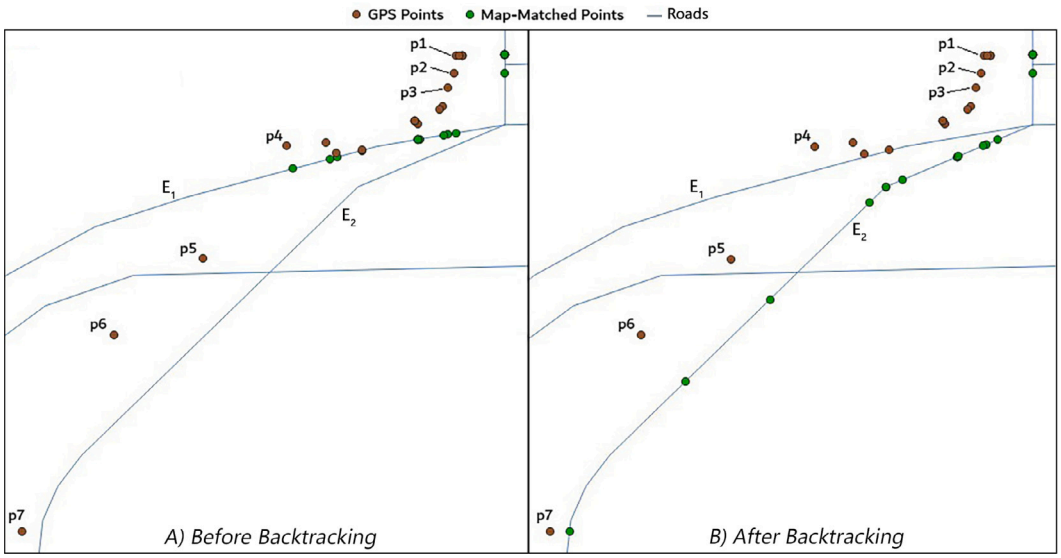

Fig. 11. Result of Map-Matching Using Backtracking Technique.

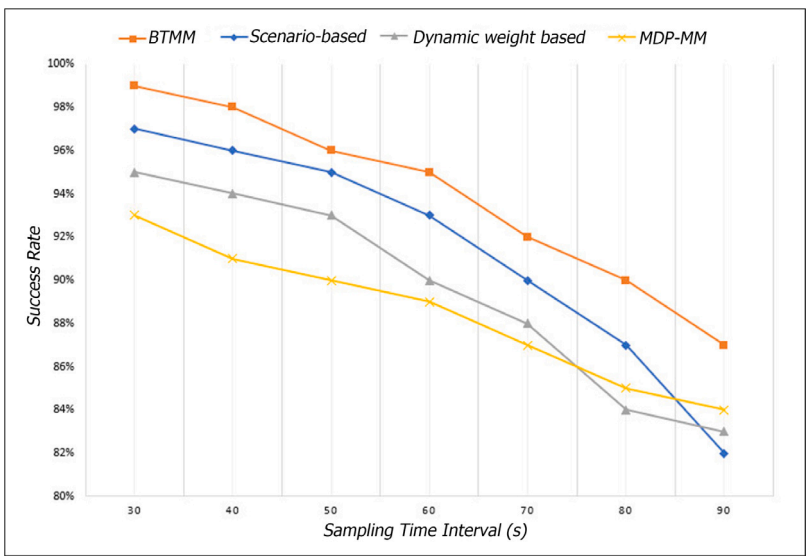

Fig. 12. Comparison of Success Rates.

As illustrated in Fig. 14, the dynamic buffer size approach achieves the highest computational speed of 8 points per second, which is superior to static buffer sizes. This demonstrates that the adaptive nature of the dynamic buffer significantly enhances computational efficiency while maintaining consistent accuracy due to the backtracking technique.

An important distinction of BTMM is its speed-independent approach. In contrast to other methodologies that use speed attribute, the BTMM approach demonstrates efficient operation without depending on speed data. This particular feature enhances the adaptability of BTMM, particularly for datasets that do not include speed information. In this particular experiment, the dataset mostly consists of essential location data (latitude and longitude), BTMM proves to be highly compatible.

Compared to other algorithms that take into account road kinds such as main and auxiliary during its process, the BTMM technique employs a more straightforward approach by excluding road type verification. This simplicity in the map matching process results in a reduction in complexity related to the classification of road types. Consequently, the BTMM provides a very effective approach for achieving trajectory map matching process.

In contrast to some map matching methods which discard GPS points with speeds less than a certain threshold (e.g., 3 m/s), BTMM does not remove any points based on speed. This approach ensures that all available data points are utilised for map matching, making BTMM suitable for a wide range of applications, including the calculation of traffic speed. The inclusion of low-speed GPS points enhances the comprehensiveness of the trajectory data and offers useful insights for further analysis.

In summary, the strength of our BTMM algorithm, which outperforms baseline methods in terms of success rate and computational speed, can be attributed to its novel utilisation of the backtracking technique, adaptive adjustment of buffer size, independence from speed attributes, and its straightforward approach to road type handling. The combined improvements mentioned lead to the development of accuracy and the increased applicability of BTMM in various datasets and scenarios. Furthermore, the inclusion of low-speed GPS locations serves

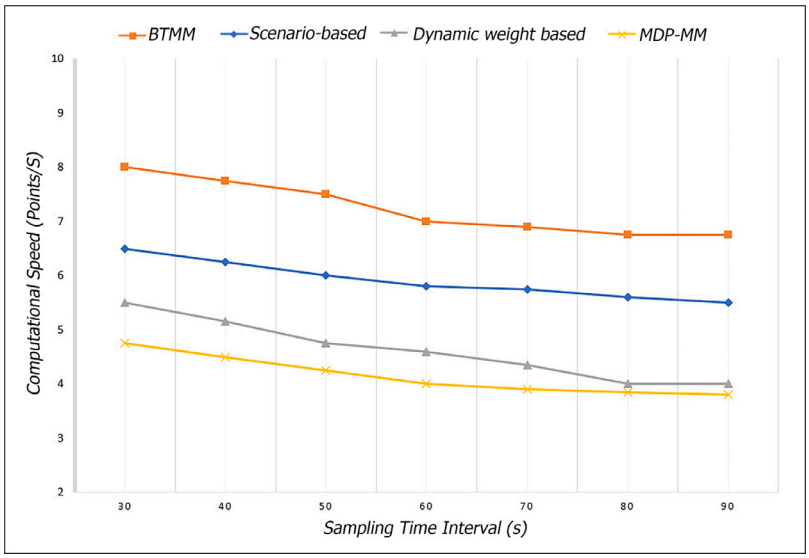

Fig. 13. Comparison of Computational Speed.

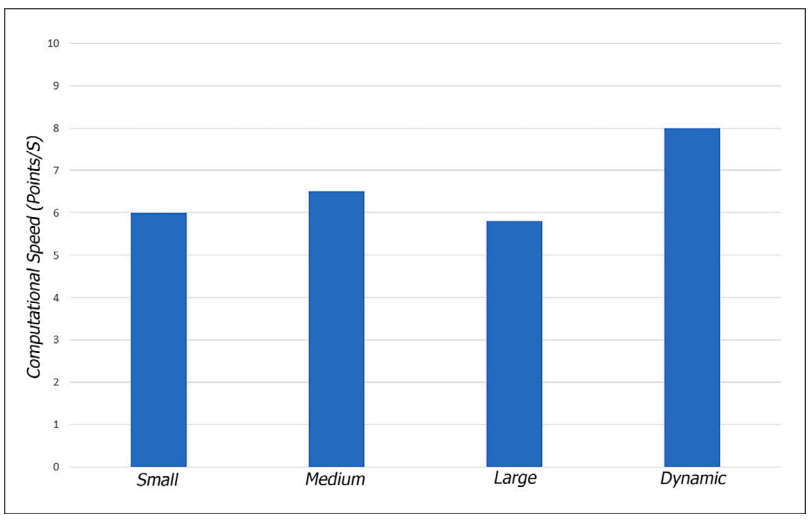

Fig. 14. Computational Speed for Different Buffer Size Strategies.

to improve the usefulness and comprehensiveness of BTMM trajectory data, making it a flexible option for a wide range of applications.

5. Conclusions

In this study, we have introduced a novel map matching approach that combines the power of topological information and integrates the backtracking technique to achieve a high level of accuracy in aligning observed trajectories with road networks, called Backtracking Technique Map Matching (BTMM). Our method has demonstrated its effectiveness through comprehensive testing on real-world datasets, achieving an impressive accuracy rate of up to 99.92

The innovative fusion of topological information with backtracking technique has greatly improved the success rate of our map matching algorithm. By using the underlying logical relationships within road networks, our approach has overcome challenges posed by complex road geometries and urban environments, where traditional geometric-based methods often falter.

However, it is important to acknowledge that the algorithm performance linked to the level of topological connection shown by the digital road network. The inadequate digitisation of road network maps can pose limitations to the algorithm ability to use topological relationships effectively, potentially leading to incorrect results.

In order to overcome this limitation, future work will prioritise the implementation of additional calibration procedures for our method. Our objective is to investigate approaches that may improve the algorithm capacity to adapt to different levels of digitisation quality. We intend to enhance the effectiveness and precision of our technique by adjusting the algorithm responsiveness to topological information. This adjustment is to minimise the negative effects caused by inadequately digitised road networks, therefore expanding the range of situations in which our approach may be used and improving its accuracy.

In anticipation of future inquiries, our study paves the way for more studies aimed at enhancing and modifying our approach to suit diverse circumstances. Addressing the challenges posed by incomplete road network digitisation remains a crucial topic of focus for future research. Additionally, the promising accuracy achieved by our approach prompts further investigation into potential applications beyond map matching, including but not limited to urban planning, transportation optimisation, and location-based services.

CRediT authorship contribution statement

Hussein Alkhazraji: Writing – original draft, Software, Project administration, Methodology, Investigation. **Triantafyllos Kanakis:** Writing – review & editing, Supervision. **Ali Al-Sherbaz:** Writing – review & editing, Supervision. **Wesam Sameer Bhaya:** Writing – review & editing, Supervision. **Michael Opoku Agyeman:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data underpinning this publication are openly available from the Microsoft website at https://www.microsoft.com/en-us/research/publication/t-drive-trajectory-data-sample/.

Acknowledgements

This research was funded by the Iraqi ministry of higher education and scientific research, Republic of IRAQ (Ref. [22219]) in (06/09/2017) according to the scholarship to sponsor the first author to pursue his PhD research.

References

Bernstein, D., Kornhauser, A., et al., 1996. An introduction to map matching for personal navigation assistants.

Chao, P., Xu, Y., Hua, W., Zhou, X., 2020. A survey on map-matching algorithms. In: Databases Theory and Applications: 31st Australasian Database Conference, ADC 2020, Melbourne, VIC, Australia, February 3–7, 2020, Proceedings 31. Springer, pp. 121–133.

Chen, B.Y., Yuan, H., Li, Q., Lam, W.H., Shaw, S.L., Yan, K., 2014. Map-matching algorithm for large-scale low-frequency floating car data. Int. J. Geogr. Inf. Sci. 28 (1), 22–38.

De Fabritiis, C., Ragona, R., Valenti, G., 2008. Traffic estimation and prediction based on real time floating car data. In: 2008 11th International IEEE Conference on Intelligent Transportation Systems. IEEE, pp. 197–203.

Dia, H., Thomas, K., 2011. Development and evaluation of arterial incident detection models using fusion of simulated probe vehicle and loop detector data. Inf. Fusion 12 (1), 20–27.

Gao, C., Xu, J., Li, Q., Yang, J., 2019. The effect of posted speed limit on the dispersion of traffic flow speed. Sustainability 11 (13), 3594.

Gurari, E., 1999. CIS 680: Data structures: Chapter 19: Backtracking algorithms. p. 5632, Retrieved from Wayback Machine: https://web.archive.org/web/20070.31701.

Haoyan, W., Yuangang, L., Shaohua, L., Bo, L., Zongyi, H., 2023. A path increment map matching method for high-frequency trajectory. IEEE Trans. Intell. Transp. Syst..

Hashemi, M., Karimi, H.A., 2016. A weight-based map-matching algorithm for vehicle navigation in complex urban networks. J. Intell. Transp. Syst. 20 (6), 573–590.

Hoboken, N.W., 2006. Planning and urban design standards.

Jabbour, M., Bonnifait, P., Cherfaoui, V., 2008. Map-matching integrity using multihypothesis road-tracking. J. Intell. Transp. Syst. 12 (4), 189–201.

Kong, X., Yang, J., 2019. A scenario-based map-matching algorithm for complex urban road network. J. Intell. Transp. Syst. 23 (6), 617–631.

Kong, X., Yang, J., Yang, Z., 2015. Measuring traffic congestion with taxi GPS data and travel time index. In: CICTP 2015. pp. 3751–3762.

Kubicka, M., Cela, A., Mounier, H., Niculescu, S.I., 2018. Comparative study and application-oriented classification of vehicular map-matching methods. IEEE Intell. Transp. Syst. Mag. 10 (2), 150–166.

Li, X., Ma, S., Yang, H., Zhang, X., 2017. Vector road matching algorithm based on a topological path. J. Image Graph 22 (05), 596–609.

Liu, M., Zhang, L., Ge, J., Long, Y., Che, W., 2020. Map matching for urban high-sampling-frequency GPS trajectories. ISPRS Int. J. Geo-Inf. 9 (1), 31.

Newson, P., Krumm, J., 2009. Hidden Markov map matching through noise and sparseness. In: Proceedings of the 17th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems. pp. 336–343.

Pan, G., Qi, G., Wu, Z., Zhang, D., Li, S., 2012. Land-use classification using taxi GPS traces. IEEE Trans. Intell. Transp. Syst. 14 (1), 113–123.

Quddus, M.A., Ochieng, W.Y., Noland, R.B., 2007. Current map-matching algorithms for transport applications: State-of-the art and future research directions. Transp. Res. Part C: Emerg. Tech. 15 (5), 312–328.

Sanaullah, I., Quddus, M., Enoch, M., 2016. Developing travel time estimation methods using sparse GPS data. J. Intell. Transp. Syst. 20 (6), 532–544.

Satrinia, D., Saptawati, G.P., 2017. Traffic speed prediction from GPS data of taxi trip using support vector regression. In: 2017 International Conference on Data and Software Engineering. ICoDSE, IEEE, pp. 1–6.

Schuessler, N., Axhausen, K.W., 2009. Map-matching of GPS traces on high-resolution navigation networks using the multiple hypothesis technique (MHT). Arbeitsberichte Verkehrs-und Raumplanung 568.

Velaga, N.R., Quddus, M.A., Bristow, A.L., 2009. Developing an enhanced weight-based topological map-matching algorithm for intelligent transport systems. Transp. Res. C 17 (6), 672–683.

Velaga, N.R., Quddus, M.A., Bristow, A.L., 2012. Improving the performance of a topological map-matching algorithm through error detection and correction. J. Intell. Transp. Syst. 16 (3), 147–158.

Wang, Z., Goodchild, A., McCormack, E., 2016. Measuring truck travel time reliability using truck probe GPS data. J. Intell. Transp. Syst. 20 (2), 103–112.

Washington, D.C., 2008. Global positioning system standard positioning service performance standard.

Weber, M., Liu, L., Jones, K., Covington, M.J., Nachman, L., Pesti, P., 2010. On map matching of wireless positioning data: A selective look-ahead approach. In: Proceedings of the 18th SIGSPATIAL International Conference on Advances in Geographic Information Systems. pp. 290–299.

Wei, H., Wang, Y., Forman, G., Zhu, Y., 2013. Map matching by Fréchet distance and global weight optimization. Tech. Pap., Dep. Comput. Sci. Eng. 19.

Xu, H., Liu, H., Tan, C.-W., Bao, Y., 2010. Development and application of an enhanced Kalman filter and global positioning system error-correction approach for improved map-matching. J. Intell. Transp. Syst. 14 (1), 27–36.

Yang, D., Zhang, T., Li, J., Lian, X., 2011. Synthetic fuzzy evaluation method of trajectory similarity in map-matching. J. Intell. Transp. Syst. 15 (4), 193–204.

YIN, J., WU, F., LI, A., DU, J., LIU, C., 2021. Trajectory matching considering speed and heading information 46 (11). pp. 1774–1781. http://dx.doi.org/10.13203/j.whugis20210058.

Yu, Q., Hu, F., Ye, Z., Chen, C., Sun, L., Luo, Y., 2022. High-frequency trajectory map matching algorithm based on road network topology. IEEE Trans. Intell. Transp. Syst. 23 (10), 17530–17545.

Zhao, X., Cheng, X., Zhou, J., Xu, Z., Dey, N., Ashour, A.S., Satapathy, S.C., 2018. Advanced topological map matching algorithm based on D–S theory. Arab. J. Sci. Eng. 43, 3863–3874.

Zheng, Y., 2011. T-drive trajectory data sample. URL https://www.microsoft.com/en-us/research/publication/t-drive-trajectory-data-sample/.

Zhu, D., Liu, Y., 2017. An incremental map-matching method based on road network topology 42 (1). pp. 77–83. http://dx.doi.org/10.13203/j.whugis20150016.